

BSR Transition Risk Layer

Methodology Reference — Implementation v1 (Session 8)

This document specifies the four-layer transition risk model implemented in `engine/transition/` and surfaced via `pages/11_TransitionRisk.py`. It is the canonical reference for the methodology, calibration choices, and known limitations of the BSR Climate Risk Intelligence Platform's transition module.

Reference architecture: BSR Climate Risk Practice memo “*Quantifying Transition Risk: Landscape, Frontier, and a Path Forward*” (internal, May 2026). The four layers below productise Path A of that memo: a reusable Python toolkit for sector-specific, channel-decomposed, DCF-integrable transition risk analysis.

1. Overview

The transition risk module decomposes the financial impact of decarbonisation on an asset (or a portfolio of assets) into four channels. Each channel maps to a specific empirical or theoretical literature, routes to exactly one financial destination (cash flow, asset value, or discount rate), and is independently calibratable. The four channels can be enabled or disabled individually.

Layer	Channel	Mechanism	Routes to
L1	Carbon cost	Scope 1+2 emissions × NGFS scenario carbon price × (1 – pass-through)	Cash flow OpEx
L2	Technology / stranding	Wright's Law cost crossover (or sector demand collapse) → logistic impairment	Cash flow + asset value
L3	Network propagation	Sector-typical shock vector → Leontief inverse → focal sector input cost	Cash flow input cost
L4	Reputational / capital	Sautner CCEXposure × empirical elasticities (credit / equity / revenue)	Cash flow OR WACC (one only)

Non-duplication rule. Each channel routes exactly once. Layer 4 is the only layer with a configurable destination (cash flow vs WACC); the orchestrator zeros out the alternative path so the same financial impact is never counted twice. Stranded-asset impairment from Layer 2 is reported separately from the cash-flow stream — consistent with the BSR framework's distinction between operating-cost adjustment and balance-sheet impairment.

2. Layer 1 — Direct Carbon Cost with Pass-Through

2.1 Mathematics

For each (scenario, region, year, sector), the layer computes a four-way decomposition of the firm's carbon cost:

```
gross_cost = (scope1 + scope2) × carbon_price
absorbed = gross_cost × (1 – pass_through) ← margin compression
passed_through = gross_cost × pass_through ← becomes Layer-3 shock
scope3_indirect = scope3 × carbon_price × (1 – pass_through)
net_carbon_opex = absorbed + scope3_indirect ← what hits the firm CFs
```

The pass-through coefficient encodes the empirical observation that downstream price elasticity, supply elasticity, market structure, and trade exposure determine how much of a carbon cost can be passed to customers. The remainder compresses producer margins. Two firms with identical Scope 1+2 emissions in the same scenario can experience very different financial impacts depending on these structural parameters.

2.2 Data

Carbon prices — NGFS Phase V REMIND-MagPIE 3.2 outputs (Nov 2023), regionalised into three groups (advanced, emerging, rest-of-world) consistent with the REMIND model regions. Stored in `data/transition/carbon_prices_ngfs.json`. IPCC SSP scenarios fall back to their closest NGFS analog.

Pass-through coefficients — Sector-median values from the published empirical literature on emissions cost incidence. Stored in `data/transition/sector_pass_through.json` with per-sector demand elasticity, market structure tag, and trade-exposure flag.

2.3 Sources

- Sijm, Hers, Lise, Wetzelaer (2012). “CO2 Cost Pass-Through and Windfall Profits in the Power Sector.” *Energy Policy*. — Power 60–100% pass-through.
- Fabra & Reguant (2014). “Pass-Through of Emissions Costs in Electricity Markets.” *American Economic Review* 104(9). — Empirical merit-order test.
- Cludius, Hermann, Matthes, Graichen (2020). “The merit order effect of wind and photovoltaic electricity generation in Germany 2008-2016.” *Energy Economics*.
- Frankovic (2022). “The impact of carbon pricing in a multi-region production network model.” Bundesbank Discussion Paper 06/2022.
- Devulder & Lisack (2020). “Carbon tax in a production network.” Banque de France WP #813.

3. Layer 2 — Technology Disruption (Learning Curves)

3.1 Wright's Law

Each technology has a learning rate (LR), the fractional cost reduction per doubling of cumulative installed capacity. Cost evolves as:

$$\text{Cost}(Q) = \text{Cost}(Q_{\text{ref}}) \times (Q / Q_{\text{ref}})^{-b}$$

where $b = \log_2(1 - \text{LR})$

Cumulative capacity index Q/Q_{ref} is computed from a scenario-specific compound growth rate. Negative growth values in the data (representing demand decline) are clamped to zero — cumulative capacity is monotonic non-decreasing; demand-side decline is captured via the sector pathway, not the technology cost curve.

3.2 Lafond Distributional Forecast

The point estimate above is wrapped in a log-normal forecast band per Lafond et al. (2018). One-sigma bounds at horizon t are:

$$\log \text{Cost}_t \sim \text{Normal}(\mu = \log \text{Cost}_{\text{pred}}(t), \sigma = \text{lafond}_\sigma \times \sqrt{t})$$

$$\text{cost}_{\text{lo}} = \text{pred} \times \exp(-\sigma_t) \quad \text{cost}_{\text{hi}} = \text{pred} \times \exp(+\sigma_t)$$

lafond_σ values are calibrated from the Santa Fe Performance Curve Database. Mature technologies (solar PV, lithium-ion batteries) have tighter bands than emerging ones (electrolysers, H2-DRI steel).

3.3 Stranding Trigger

An asset is flagged for stranding when one of two triggers fires within the analysis horizon:

- **Cost crossover** — challenger technology's projected cost $\leq$ incumbent's. Detected by year-by-year comparison along the Wright's Law trajectories.
- **Demand collapse** (fossil-dependent sectors only) — sector pathway falls below 0.5 (50% demand reduction relative to 2025 baseline). Captures stranding from demand decline alone, e.g. refineries when oil demand falls regardless of biorefining cost.

Once triggered, impairment follows a logistic curve centred on the trigger year:

```
frac(y) = 1 / (1 + exp(-slope * (y - y_crossover))) where slope = 0.20
annual_impairment(y) = (frac(y) - frac(y-1)) * cap
cap = max(0, 1 - pathway(2050)) * replacement_value * λ
λ = 1.0 if fossil_dependent else 0.5
```

The slope of 0.20 produces ~50% impairment at the trigger year and ~95% completion after 15 years — empirically consistent with typical fossil-asset response times once a challenger reaches cost parity. The cap scales by the scenario-specific demand collapse so the same trigger year produces different stranded shares under different scenarios. Non-fossil sectors cap at 50% of replacement value (only the share tied to the incumbent technology can strand).

3.4 Sources

- Way, Ives, Mealy, Farmer (2022). “Empirically grounded technology forecasts and the energy transition.” *Joule* 6(9). — Canonical Wright's Law application to NGFS.
- Lafond, Bailey, Bakker, Rebois, Zadourian, McSharry, Farmer (2018). “How well do experience curves predict technological progress?” *TFSC* 128. — Distributional forecast methodology.
- Ziegler & Trancik (2021). “Re-examining rates of lithium-ion battery technology improvement and cost decline.” *Energy & Environmental Science* 14.
- IRENA (2023). *Renewable Power Generation Costs in 2022*.

4. Layer 3 — Production Network Propagation

4.1 Leontief Inverse

Sectoral cost shocks propagate through input-output linkages. Given the direct requirements matrix A — element $A[i,j]$ = USD of input from sector i required to produce 1 USD of sector j 's output — the Leontief inverse $L = (I - A)^{-1}$ aggregates all rounds of indirect input requirements.

Under a Cobb-Douglas production assumption (substitution elasticity $\sigma = 1$), the total cost shock absorbed by sector j is:

```
total_shock_j = Σ L[i, j] * s_i
where s_i = passed-through cost in sector i (fraction of sector i's output)
```

4.2 Sectoral Shock Vector

For a per-asset analysis, the shock vector is built from sector-typical exposure rather than the focal asset's own pass-through (which would double-count). For each sector i :

```
s_i = emission_intensity_i * carbon_price * pass_through_i * 10^9
(emission_intensity is in tCO2e per million USD revenue; the 10^9 converts to per-USD)
```

The focal asset's indirect cost is then propagated $\times$ asset_revenue, less the asset's own diagonal contribution (which is already captured by Layer 1). The output also lists the top-5 upstream sectors by contribution — useful for supply-chain attribution.

4.3 Substitution Elasticity (CES Damping)

The default Cobb-Douglas case ($\sigma=1$) is conservative but rigid. The implementation supports a CES-style damping: off-diagonal entries of A are scaled by $1/\sigma$ before inversion, capturing the empirical observation that intermediate-input substitution moderates shock propagation. Empirical estimates cluster $\sigma \in [1.3, 3.0]$ (Papageorgiou et al. 2017).

4.4 Limitations

The shipped I-O matrix is a 20-sector single-region aggregation calibrated against EXIOBASE-3 world totals (2019). For production use — the bespoke add-on path in the memo — replace with the full multi-regional MRIO (200 sectors $\times$ 49 regions). The current formulation also does not include endogenous default thresholds; Reisch et al. (2025) is the natural extension.

4.5 Sources

- Stadler, Wood, Bulavskaya et al. (2018). “EXIOBASE 3.” *Journal of Industrial Ecology* 22(3). — Underlying I-O calibration.
- Acemoglu, Carvalho, Ozdaglar, Tahbaz-Salehi (2012). “The Network Origins of Aggregate Fluctuations.” *Econometrica* 80(5). — Theoretical foundation.
- Reisch, Diem, Pichler, Stangl, Thurner (2025). “Combined climate stress testing of supply-chain networks and the financial system.” arXiv:2503.10644 — Endogenous-default extension.
- Papageorgiou, Saam, Schulte (2017). “Substitution between Clean and Dirty Energy Inputs.” *REStat* 99(2). — Substitution elasticity calibration.

5. Layer 4 — Reputational and Capital-Access Overlay

5.1 CCExposure Decomposition

Sautner, van Lent, Vilkov, Zhang (2023) measure firm-level Climate Change Exposure from earnings-call transcripts and decompose it into three sub-measures:

- **Opportunity** — positive narrative; revenue-uplift signal.
- **Regulatory** — policy / legal exposure; cost and spread signal.
- **Physical** — climate-physical-impact discussion; cost and spread signal.

The shipped proxy uses sector medians from the published 2010–2020 dataset. For production use, override `firm_override` in `compute_exposure_premium()` with the licensed firm-level Sautner data.

5.2 Empirical Elasticities

Per-unit pricing impacts from Sautner et al. (2023, *Management Science*):

```
credit_spread_abps = 12 × CCE_regulatory + 6 × CCE_physical
equity_premium_abps = 50 × (CCE_opportunity + CCE_regulatory + CCE_physical)
revenue_growth_abps = 35 × CCE_opportunity – 25 × CCE_regulatory
```

Scenario modifiers amplify or dampen the three sub-measures so that disorderly transitions widen the regulatory channel, ordered transitions widen the opportunity channel, and high-warming scenarios widen the

physical channel.

5.3 Routing Rule

Layer 4 is the only layer with a configurable destination. The routing parameter must be set explicitly to one of:

- **ROUTE_CASHFLOWS** (default) — apply the revenue growth modifier to the cash-flow stream; the WACC-side credit/equity premiums are computed for diagnostic display only.
- **ROUTE_WACC** — apply the credit + equity premium to the discount rate; revenue modifier is zeroed. Useful when modelling firm-systematic exposure rather than idiosyncratic revenue dynamics.

5.4 Sources

- Sautner, van Lent, Vilkov, Zhang (2023). “Firm-Level Climate Change Exposure.” *Journal of Finance* 78(3). — CCExposure construction.
- Sautner et al. (2023). “Pricing Climate Change Exposure.” *Management Science* (forthcoming). — Empirical elasticities.
- Engle, Giglio, Kelly, Lee, Stroebe (2020). “Hedging Climate Change News.” *RFS* 33(3). — Climate news beta as triangulation.

6. Orchestration and DCF Integration

`engine.transition.transition_engine.run_asset_transition()` computes one asset × scenario × horizon. Per-asset outputs feed the existing `engine.dcf_engine.compute_climate_dcf()` via the long-form damage DataFrame produced by `transition_results_to_damage_df()`. The combined DCF (physical + transition) is exposed as `engine.transition.transition_dcf.compute_combined_dcf()` and returns three side-by-side NPVs:

NPV	Damage stream	Use
Physical-only	EAD timeline from physical engine	Physical-risk-only DCF impairment
Transition-only	L1+L2+L3+L4 cash-flow cost stream	Transition-risk attribution
Combined	Sum of physical and transition by year	Headline climate-adjusted NPV

Stranded-asset PV impairment is computed independently (sum of `annual_impairment_usd` discounted at WACC) and reported as `total_pv_stranded_impairment` — separate from the cash-flow stream so users can isolate value impairment from operating-cost adjustment.

7. Calibration — Worked Examples

All figures from a fresh run of `run_asset_transition()` at the default settings (all four layers enabled, Layer-4 routing to cash flows, $\sigma = 1.0$). Currency is USD.

7.1 Coal-Fired Power Plant — USA

Inputs: \$500M replacement value, 2.5 MtCO₂/yr Scope 1, 50 ktCO₂/yr Scope 2, 200 ktCO₂/yr Scope 3, \$300M annual revenue, sector = power_coal.

Scenario	L1 2050	L2 rev 2050	L3 2050	L4 2050	Total CF 2050	Cum. impairment 2025–50
Net Zero 2050	\$169M	\$294M	\$0.1M	\$31.7M	\$495M	\$266M
Current Policies	\$7M	\$36M	\$0.0M	\$15.4M	\$59M	\$33M
Fragmented World	\$92M	\$ —	\$0.1M	\$ —	\$377M	\$168M

Stranding crossover is 2025 across all scenarios because solar PV is already cheaper than new-build coal. The scenario differentiation comes from (a) the Layer-1 carbon cost, which scales linearly with the regional carbon price, and (b) the Layer-2 impairment cap, which scales with the scenario-specific demand collapse (power_coal pathway: 0.02 in NZ2050, 0.88 in Current Policies).

7.2 Oil Refinery — USA

Inputs: \$2B replacement value, 4.5 MtCO₂/yr Scope 1, 0.3 MtCO₂/yr Scope 2, 8 MtCO₂/yr Scope 3, \$8B revenue, sector = oil_refining.

Scenario	L1 2050	L2 rev 2050	Total CF 2050	Cum. impairment 2025–50	Crossover
Net Zero 2050	\$1.3B	\$6.4B	\$8.4B	\$1.36B	2039
Delayed Transition	\$1.2B	\$5.6B	\$8.0B	\$1.05B	2044
Current Policies	\$58M	\$640M	\$1.0B	\$0	no trigger

Refining has no cost crossover (biorefining stays more expensive through 2050), but the demand-collapse trigger fires when the oil_refining pathway crosses below 0.5 — in NZ2050 that happens around 2039. Under Current Policies, demand stays above 0.92 and no impairment is triggered. The Layer-2 revenue erosion (\$6.4B in NZ2050) is the dominant component because the firm has \$8B revenue and the pathway falls to 0.20.

7.3 Commercial Office Building — USA

Inputs: \$50M replacement value, 200 tCO₂/yr Scope 1, 800 tCO₂/yr Scope 2, \$15M revenue, sector = real_estate_commercial.

Scenario	L1 2050	L4 2050	Total CF 2050	Cum. impairment
Net Zero 2050	\$0.29M	–\$0.52M	–\$0.22M	\$0
Current Policies	\$0.01M	–\$0.38M	–\$0.36M	\$0

An office building with a low-carbon footprint shows a small net *positive* transition outcome under both scenarios — the Layer-4 opportunity premium (real_estate_commercial CCE opportunity = 0.85; growth bps = $+35 \times 0.85 - 25 \times 0.95 \approx -0.50$ after scenario modifiers, but cumulative compounding pushes the dollar

revenue uplift ahead of the modest Layer-1 carbon cost). No stranding because real estate has no fossil-dependent flag and the sector pathway is mildly positive in all scenarios.

8. Known Limitations

- **I-O resolution.** The shipped 20-sector single-region matrix is a screening tool. Production deployments should swap in EXIOBASE-3 MRIO (200 × 49) — the bespoke add-on path in the memo. The Leontief solver in `network_propagation.py` works on any square matrix; only the data file changes.
- **CCExposure data.** Sector medians stand in for firm-level scores. The firm-level Sautner et al. dataset is licensed separately; pass it through `firm_override`.
- **Carbon-price snapshot.** Values approximate published NGFS Phase V REMIND-MAgPIE outputs as of November 2023. Refresh from the NGFS Scenarios Portal for regulatory disclosures — phase updates change values materially.
- **Pass-through.** Sector-median values; firm-specific values vary materially with market power, contract structure, and trade exposure. Override per-asset via `pass_through_override`.
- **No endogenous default.** Layer 3 propagates linearly through the IO network; the Reisch et al. (2025) endogenous-default extension is not implemented.
- **Scope 3 simplification.** Scope 3 is treated as a single number absorbed by the $(1 - \text{pass_through})$ coefficient. A full upstream Scope 3 decomposition would be category-by-category (purchased goods, transport, use-of-sold-products) and route through Layer 3 explicitly.

9. Implementation Pointers

Component	Path
Orchestrator entry	<code>engine.transition.transition_engine.run_asset_transition</code>
Portfolio orchestrator	<code>engine.transition.transition_engine.run_portfolio_transition</code>
DCF integration	<code>engine.transition.transition_dcf.compute_combined_dcf</code>
Layer 1	<code>engine.transition.carbon_pricing.compute_carbon_cost</code>
Layer 2	<code>engine.transition.learning_curves.compute_stranding</code>
Layer 3	<code>engine.transition.network_propagation.propagate_carbon_shock</code>
Layer 4	<code>engine.transition.cc_exposure.compute_exposure_premium</code>
Streamlit UI	<code>pages/11_TransitionRisk.py</code>
Test suite	<code>tests/test_transition.py</code> (42 tests)
Carbon prices	<code>data/transition/carbon_prices_ngfs.json</code>
Pass-through table	<code>data/transition/sector_pass_through.json</code>
Learning curves	<code>data/transition/learning_curves.json</code>
Sector pathways	<code>data/transition/sector_pathways.json</code>
CCExposure proxy	<code>data/transition/cc_exposure_proxy.json</code>
I-O matrix	<code>data/transition/io_matrix.json</code>
Sector taxonomy	<code>data/transition/sector_taxonomy.json</code>

Appendix A — Sector Taxonomy

Sectors implemented in the transition module. Emission intensity is in tCO₂e per million USD of revenue (sector-typical).

Key	Label	Incumbent tech	Challenger tech	Fossil-dep.	EI (t/M\$)
power_coal	Electric Power — Coal	coal_steam	solar_pv	✓	7.50
power_gas	Electric Power — Natural Gas	gas_ccgt	battery_storage	✓	3.00
power_renewable	Electric Power — Renewables	solar_pv	—	—	0.05
oil_upstream	Oil & Gas — Upstream	oil_extraction	—	✓	1.20
oil_refining	Oil Refining & Marketing	oil_refining	biorefining	✓	0.55
gas_distribution	Natural Gas Distribution	gas_distribution	heat_pump	✓	0.40
steel	Steel & Iron	blast_furnace	h2_dri	✓	1.85
cement	Cement & Lime	cement_clinker	low_clinker_ccus	✓	1.60
chemicals	Chemicals (Bulk & Petrochemicals)	petrochemical	green_h2	✓	0.95
aluminium	Aluminium & Non-Ferrous Metals	smelter	renewable_smelter	✓	1.40
road_transport_ice	Automotive — Internal Combustion	ice_vehicle	ev	✓	0.15
road_transport_ev	Automotive — Electric Vehicles	ev	—	—	0.10
aviation	Aviation (Passenger & Freight)	jet_fuel	saf	✓	0.70
shipping	Shipping & Maritime	bunker_fuel	ammonia	✓	0.80
real_estate_commercial	Real Estate — Commercial	gas_boiler	heat_pump	—	0.08
real_estate_residential	Real Estate — Residential	gas_boiler	heat_pump	—	0.06
agriculture	Agriculture & Livestock	intensive_livestock	regenerative	—	0.60
data_center	Data Centres	grid_electricity	renewable_ppa	—	0.40
manufacturing_general	General Manufacturing	mixed_industrial	—	—	0.25
services	Services & Other	office_operations	—	—	0.05

Appendix B — Pass-Through Coefficients

Sector-median values drawn from the published empirical literature. Demand elasticity is short-run own-price (Marshallian).

Sector	PT	Demand - η	Market structure	Trad e	Source
power_coal	85%	-0.20	regulated_oligopoly	—	Sijm 2012; Fabra & Reguant 2014
power_gas	80%	-0.20	regulated_oligopoly	—	Sijm 2012
power_renewable	75%	-0.20	regulated_oligopoly	—	Cludius 2020 (merit-order effect)
oil_upstream	40%	-0.30	global_commodity	✓	Frankovic 2022 (low pass-through, global price taker)
oil_refining	75%	-0.25	regional_oligopoly	✓	Sijm 2012 (refining ~70-80%)
gas_distribution	90%	-0.30	regulated_monopoly	—	Regulated cost recovery
steel	30%	-0.50	global_competitive	✓	Fabra & Reguant 2014; CEMBUREAU/EUROFER carbon-leakage submissions
cement	50%	-0.40	regional_oligopoly	✓	EU ETS sector analyses; Pang & Brown 2017
chemicals	40%	-0.45	global_competitive	✓	Devulder & Lisack 2020
aluminium	25%	-0.55	global_competitive	✓	EU CBAM impact assessment 2021
road_transport_ice	65%	-0.35	regional_oligopoly	✓	Goolsbee 1998 (auto pricing); Knittel & Tanaka 2015
road_transport_ev	60%	-0.50	regional_oligopoly	✓	EV demand elasticity from RAND 2021
aviation	55%	-0.80	regional_oligopoly	✓	InterVISTAS 2007 (price elasticity); ICAO 2022
shipping	70%	-0.20	global_competitive	✓	Faber et al. 2017 IMO study
real_estate_commercial	30%	-0.20	local_competitive	—	Eichholtz, Kok, Quigley 2010 (green-building rents)
real_estate_residential	20%	-0.15	local_competitive	—	Aydin, Brounen, Kok 2020 (residential energy capitalisation)
agriculture	30%	-0.40	global_competitive	✓	Hertel et al. 2022 (GTAP-AGR)
data_center	50%	-0.30	regional_competitive	—	Hyperscaler PPA structures (AWS/MSFT 2023 CDP filings)
manufacturing_general	40%	-0.40	regional_competitive	✓	Devulder & Lisack 2020 (industrial average)
services	60%	-0.30	local_competitive	—	Default (low intensity, local pricing power)

Appendix C — Technology Learning Rates

Learning rate = fractional cost reduction per doubling of cumulative installed capacity. Lafond σ controls the width of the log-normal forecast band.

Challenger technologies

Technology	LR	Lafond σ	Source
Solar PV (utility-scale)	24.0%	0.085	IRENA 2023; Way et al. 2022 Table S1
Onshore Wind	19.0%	0.075	IRENA 2023
Offshore Wind	21.0%	0.110	IRENA 2023; Way et al. 2022
Lithium-ion Battery Storage	22.0%	0.090	Ziegler & Trancik 2021
Electric Vehicle (battery + powertrain)	18.0%	0.070	Way et al. 2022; BNEF EV Outlook 2023
Hydrogen Electrolyser (PEM/ALK)	16.0%	0.130	IEA Global Hydrogen Review 2023
Heat Pump (residential / commercial)	11.0%	0.060	IEA Future of Heat Pumps 2022
Hydrogen Direct-Reduced Iron (steel)	10.0%	0.140	IEA Iron & Steel Technology Roadmap 2020
Sustainable Aviation Fuel	12.0%	0.120	ICAO LTAG 2022; IEA Aviation 2023
Green Ammonia (shipping fuel)	13.0%	0.130	IRENA & Methanol Institute 2021
Advanced Biofuel Refining	8.0%	0.100	IEA Bioenergy 2023
Low-Clinker Cement with CCUS	7.0%	0.110	IEA Cement Technology Roadmap 2018; CEMBUREAU 2050 Roadmap
Renewable-Powered Smelter	10.0%	0.090	Mission Possible Partnership 2022
Regenerative Agriculture Practices	5.0%	0.100	FAO 2022; Project Drawdown
Corporate Renewable PPA	20.0%	0.070	BNEF Corporate Energy Procurement 2023

Incumbent technologies

Technology	LR	Lafond σ	Source
Coal Steam Power (incumbent)	0.0%	0.040	IEA Coal 2022
Gas Combined Cycle (incumbent)	4.0%	0.040	IEA Gas Market Report 2023
Internal Combustion Vehicle (incumbent)	2.0%	0.040	IEA Global EV Outlook 2023
Conventional Oil Extraction (incumbent)	0.0%	0.050	IEA WEO 2023; Carbon Tracker 2022
Conventional Oil Refining (incumbent)	2.0%	0.040	IEA WEO 2023
BF/BOF Steel (incumbent)	2.0%	0.040	WorldSteel 2023
Conventional Clinker Cement (incumbent)	1.0%	0.040	CEMBUREAU 2050 Roadmap
Petrochemical Feedstock (incumbent)	3.0%	0.050	IEA Petrochemicals 2018

Technology	LR	Lafond σ	Source
Conventional Aluminium Smelter (incumbent)	4.0%	0.050	International Aluminium Institute 2022
Jet Kerosene (incumbent)	1.0%	0.040	IEA Aviation 2023
Marine Bunker Fuel (incumbent)	1.0%	0.040	IMO 2023 GHG Strategy
Gas Boiler (buildings, incumbent)	2.0%	0.040	IEA Future of Heat Pumps 2022
Gas Distribution Network (incumbent)	1.0%	0.040	IEA Gas 2023
Intensive Livestock (incumbent)	2.0%	0.050	FAO Livestock 2022
Grid Electricity (mixed source)	10.0%	0.050	IEA WEO 2023 (electricity demand)
Office Operations (incumbent)	3.0%	0.050	Default services baseline
Mixed Industrial Operations	4.0%	0.050	Industrial average

Appendix D — NGFS Carbon Prices (USD/tCO₂e)

NGFS Phase V REMIND-MAgPIE 3.2 carbon-price trajectories, sampled at 5-year intervals. Region groups: **advanced** (USA, EU, JPN, AUS, CAN, KOR, SGP, ISR), **emerging** (CHN, IND, BRA, MEX, IDN, RUS, ZAF, TUR...), **rest-of-world** (default).

Region: Advanced

Scenario	2025	2030	2035	2040	2045	2050
Net Zero 2050	\$35	\$130	\$195	\$260	\$335	\$410
Below 2C	\$20	\$55	\$95	\$135	\$185	\$240
Divergent Net Zero	\$40	\$145	\$215	\$285	\$360	\$430
Delayed Transition	\$5	\$10	\$90	\$230	\$295	\$360
NDCs Only	\$10	\$17	\$22	\$30	\$38	\$45
Current Policies	\$5	\$8	\$10	\$12	\$15	\$18
Fragmented World	\$25	\$85	\$150	\$220	\$285	\$350
IEA NZE	\$40	\$140	\$200	\$250	\$230	\$250
IEA APS	\$30	\$95	\$130	\$165	\$175	\$175
IEA STEPS	\$15	\$35	\$45	\$55	\$65	\$75

Region: Emerging

Scenario	2025	2030	2035	2040	2045	2050
Net Zero 2050	\$20	\$85	\$140	\$195	\$260	\$330
Below 2C	\$10	\$35	\$65	\$95	\$135	\$180
Divergent Net Zero	\$15	\$70	\$120	\$175	\$240	\$310
Delayed Transition	\$3	\$5	\$60	\$165	\$220	\$280
NDCs Only	\$3	\$8	\$12	\$18	\$24	\$30
Current Policies	\$1	\$2	\$3	\$4	\$5	\$7
Fragmented World	\$3	\$10	\$20	\$35	\$55	\$80
IEA NZE	\$20	\$90	\$135	\$200	\$200	\$200
IEA APS	\$10	\$50	\$70	\$95	\$110	\$125
IEA STEPS	\$3	\$8	\$12	\$18	\$25	\$35

Region: Rest Of World

Scenario	2025	2030	2035	2040	2045	2050
Net Zero 2050	\$10	\$55	\$95	\$140	\$200	\$270
Below 2C	\$5	\$20	\$40	\$65	\$95	\$130
Divergent Net Zero	\$8	\$40	\$80	\$125	\$185	\$250
Delayed Transition	\$1	\$3	\$40	\$115	\$165	\$220
NDCs Only	\$1	\$3	\$5	\$8	\$12	\$18
Current Policies	\$0	\$1	\$1	\$2	\$3	\$4

Scenario	2025	2030	2035	2040	2045	2050
Fragmented World	\$0	\$2	\$5	\$8	\$12	\$18
IEA NZE	\$10	\$55	\$90	\$160	\$175	\$175
IEA APS	\$5	\$20	\$35	\$55	\$70	\$90
IEA STEPS	\$1	\$2	\$4	\$6	\$10	\$15

Appendix E — Bibliography

Architecture and framework

- BSR Climate Risk Practice (May 2026). *Quantifying Transition Risk: Landscape, Frontier, and a Path Forward*. Internal memo.
- BSR. *From Climate Science to Corporate Strategy*.
https://www.bsr.org/reports/BSR_Climate_Science_Corporate_Strategy.pdf

Pass-through (Layer 1)

- Sijm, Hers, Lise, Wetzelaer (2012). *Energy Policy*. DOI: 10.1016/j.enpol.2007.10.027
- Fabra & Reguant (2014). *American Economic Review* 104(9). DOI: 10.1257/aer.104.9.2872
- Cludius, Hermann, Matthes, Graichen (2020). *Energy Economics*.
- Frankovic (2022). Bundesbank Discussion Paper 06/2022.
- Devulder & Lisack (2020). Banque de France WP #813.

Learning curves (Layer 2)

- Way, Ives, Mealy, Farmer (2022). *Joule* 6(9). DOI: 10.1016/j.joule.2022.08.009
- Lafond, Bailey, Bakker, Rebois, Zadourian, McSharry, Farmer (2018). *Tech Forecasting & Social Change* 128. DOI: 10.1016/j.techfore.2017.11.001
- Ziegler & Trancik (2021). *Energy & Environmental Science* 14.
- IRENA (2023). *Renewable Power Generation Costs in 2022*.

I-O networks (Layer 3)

- Stadler, Wood, Bulavskaya et al. (2018). *Journal of Industrial Ecology* 22(3). DOI: 10.1111/jiec.12715
- Acemoglu, Carvalho, Ozdaglar, Tahbaz-Salehi (2012). *Econometrica* 80(5).
- Reisch, Diem, Pichler, Stangl, Thurner (2025). arXiv:2503.10644.
- Papageorgiou, Saam, Schulte (2017). *Review of Economics & Statistics* 99(2).

CCExposure (Layer 4)

- Sautner, van Lent, Vilkov, Zhang (2023). *Journal of Finance* 78(3). DOI: 10.1111/jofi.13219
- Sautner et al. (2023). *Management Science* (forthcoming).
- Engle, Giglio, Kelly, Lee, Stroebel (2020). *RFS* 33(3).

Scenarios

- NGFS Phase V Scenarios Portal: <https://www.ngfs.net/ngfs-scenarios-portal/>
- IEA World Energy Outlook 2023: <https://www.iea.org/reports/world-energy-outlook-2023>
- IPCC AR6 WG1 SPM: <https://www.ipcc.ch/report/ar6/wg1/chapter/summary-for-policymakers/>